

AMENDMENT NO. 1 DECEMBER 2025
TO
IS 5884 : 2020 TEXTILE FLOOR COVERING — HAND TUFTED CARPETS —
SPECIFICATION
(Third Revision)

(Page 7, Annex B) — Substitute the following for the existing:

‘ANNEX B

(Clauses 5.3.1, 5.4.1 and 5.6.1)

**METHOD OF TEST TO DETERMINE PILE THICKNESS, PILE DENSITY AND TUFT
WITHDRAWAL FORCE OF TUFTED CARPETS**

B-1 ATMOSPHERIC CONDITION FOR TESTING AND CONDITIONING OF TEST SPECIMENS

B-1.1 The test specimens shall be conditioned and the test shall be carried out in one of the standard atmospheres for conditioning and testing of textiles as specified in IS 6359.

B-1.2 Prior to testing, the test specimens shall be conditioned to moisture equilibrium in the standard atmosphere. Test specimens that have been exposed to the specified atmosphere for a period of 72 h, in such a manner that all portions of the specimens are, as far as possible, directly exposed to the atmosphere, shall be deemed to have attained moisture equilibrium.

B-2 DETERMINATION OF PILE THICKNESS AND PILE DENSITY

B-2.1 Test Specimen

B-2.1.1 Cut four specimens, each at least 200 mm × 200 mm with the two sides parallel and the other two perpendiculars to the selvedge.

B-2.1.2 If there is any tendency for the edges to fray, seal them with adhesive and allow to dry. Condition the specimens to constant weight in the standard atmosphere as given in **B-1.2**.

B-2.2 Apparatus

B-2.2.1 Carpet Shearing Equipment — clipper, trimmer, band knife machine or any other suitable shearing equipment, capable of shearing the pile uniformly and closely to the backing of the carpet without causing damage to the backing material.

B-2.2.2 Carpet Thickness Tester

- a) Shall have a circular plane pressure-foot with area between 200 mm² and 1 000 mm²;
- b) Shall be capable of exerting a pressure of 20 gf/cm² normal to the plane of the specimen; and
- c) Shall be capable of measuring thickness to an accuracy of 0.1 mm over a range 25 mm.

C-2.2.3 Sharp Point Knife

B-2.2.4 Measuring Scale — graduated in millimetre

B-2.2.5 Weighting Balance — with accuracy of 10 mg

B-2.2.6 Press and Cutter — of known area of at least 10 000 mm², which may be circular or square in shape.

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B-2.3 Test Procedure

B-2.3.1 Measure the thickness of each specimen at 5 places under the standard pressure of 20 gf/cm².

B-2.3.2 Determine the mass of pile using the method given below before the edges of the pile have been sealed as given in **B-2.1.2**:

- a) Weigh each specimen to the nearest 10 mg (M_1);
- b) Measure the length and width at four places on the back of each specimen to the nearest mm;
- c) Shear the pile from the specimen. When using a band knife machine, keep on repeating the process with the roller being set progressively lower until it is as low as possible without damaging the substrate. At each setting, insert the specimen into the machine several times, each time in a different direction. After each process, brush the pile upright. When using clippers, use forward strokes in all directions. Shear as close as possible to the substrate by running the points of the comb and cutter along the substrate without digging in. Avoid plucking any tufts or damaging the substrate. It is not necessary to shear the edges of the specimen, provided that the area of 10 000 mm² in the centre is closely shorn;
- d) Brush, blow or suction clean the specimen during and/or after shearing. Continue shearing until no further significant amount of pile yarn dust appears on the shearing blades or falls away when the specimen is shaken, pile down, over a smooth surface of contrasting colour;
- e) After shearing, unless the total area of the specimen is completely shorn without damage to the substrate and can be used to determine the mass per unit area of the substrate, cut a completely shorn piece of not less than 10 000 mm² from the centre of the specimen, using the press and cutter. In either case, the substrate yarns in this area shall be undamaged, and no tufts shall have been plucked from it;
- f) Condition each area cut out of the shorn carpet specimens by laying them flat, singly, in the atmosphere for conditioning and testing described in **B-1.1**, until successive weighing at intervals of two hours shows no progressive change greater than 0.25 percent. Record the final conditioned mass of the shorn area to the nearest 10 mg (M_2);
- g) From the measurement made in (b), calculate for each specimen the average length and width, in millimetres and the area in square millimetres (A_1);
- h) Calculate the total mass per unit area $\frac{(M_1)}{(A_1)}$ of carpet for each specimen separately, in grams per square millimetre;
- j) Measure the area (A_2), in square millimetres, of each specimen of shorn carpet;
- k) For each area of shorn carpet, calculate the mass per square millimetre $\frac{(M_2)}{(A_2)}$;
- m) For each specimen, calculate the mass of shorn pile per unit area, Q_A , to the nearest gram per square metre, using the formula:

$$Q_A = 10^6 \left(\frac{M_1}{A_1} - \frac{M_2}{A_2} \right) \text{ g/m}^2$$

where

Q_A = mass of shorn pile per unit area;

$\frac{(M_1)}{(A_1)}$ = mass per mm² of carpet before shearing; and

$\frac{(M_2)}{(A_2)}$ = mass per mm² of carpet after shearing.

- n) Calculate the mean value of the shorn mass of pile above unit area of backing from the value obtained from all the specimen (M).

B-2.3.3 Measure the thickness of each shorn specimen as obtained in **B-2.3.2(d)** at 5 places under the standard pressure of 20 gf/cm².

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B-2.4 Calculation

B-2.4.1 Pile Thickness

For each specimen, determine the mean thickness in the unshorn state and the mean thickness in the shorn state. The pile thickness of each specimen shall be calculated as the difference between these two values, expressed in millimetres to the nearest 0.02 mm. The mean pile thickness for all specimens shall then be calculated and reported to the nearest 0.1 mm (t).

B-2.4.2 Pile Density

The pile density (in g/m² per mm pile thickness) shall be calculated by dividing the mean value of mass of pile above unit area of backing (*M*) by mean value of pile thickness (*t*).

B-3 DETERMINATION OF TUFT WITHDRAWAL FORCE

B-3.1 Apparatus

B-3.1.1 Tensile testing apparatus with suitable ranges and an accuracy of ± 5 percent. Full scale loads ranging from 5 N to 100 N are generally adequate.

B-3.1.2 The apparatus shall preferably operate at a constant rate of loading, however, a constant rate of traverse or extension may also be used. In all cases, the average time required to reach the maximum withdrawal force of the tufts or loops under test shall be maintained between 5 s and 10 s. For machines that would otherwise result in excessively short test times (such as those employing load cells), this requirement may be met by introducing a spring between the loading mechanism and the specimen.

B-3.1.3 The apparatus shall have a means of clamping a specimen of carpet to base plate so that it is flat in a plane perpendicular to the direction of pull upon the tuft or loop. The flat, horizontal part of the clamp, which is brought in to contact with the specimen during the testing, shall be at least 60 mm $\times$ 60 mm, shall have the circular cut out of the 12 mm radius around the tuft, or loop to be withdrawn and shall retain the specimen inside this cut-out area (see Fig. 5). The cut-out may have a throat to allow easier positioning of the specimen with the tuft grip or hook attached.

B-3.1.4 For cut pile carpets, a suitable means of its selecting and gripping one end of tuft shall be provided (for example, surgical forceps) and for loop pile carpets, a suitable hook (for example, a knitting machine needle) is require. These tufts grip are attached to the upper jaw of the tensile testing apparatus.

B-3.1.5 The load sensing mechanism of the machine shall be calibrated with the tuft grip or hook position.

B-3.2 Test Sample and Specimens

Samples shall be selected form at least 5 places across width of the carpet. Sample shall be conditioned as mentioned in **B-1.2**. Specimen should be cut (one from each sample) to a size which is convenient for clamping on the testing machine and which will allow for at least 10 tufts to be removed from each. No tufts shall be removed from within 25 mm of the edge of a specimen or of any previously withdrawn tuft.

B-3.3 Procedure

B-3.3.1 Cut Pile Carpets

Select one end of one tuft and attach the tuft grip. Clamp the specimen firmly and attach the tuft grip to the upper jaw. Set the machine in motion and withdraw the tuft completely in a direction essentially perpendicular to that of carpet specimen. Check that only one complete tuft is withdrawn, and record the maximum force.

B-3.3.2 Loop Pile Carpets

Select a suitable loop and engage the hook under the top of it. Clamp the specimen firmly and attach the hook to the upper jaw. Set the machine the motion to move the hook in a direction essentially perpendicular to that of carpet specimen and stop it as soon as slippage occurs in the bend holding one of the loop sides. Record the maximum force.

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B-3.3.3 Repeat the procedure for a minimum of 20 tufts or loops, spread evenly through available samples, taking care that these are at least 25 mm between the previously withdrawn tuft or from the edge of specimen.

B-3.4 Report

Report the average of the tests to the nearest 0.5 N as the tuft withdrawal force.

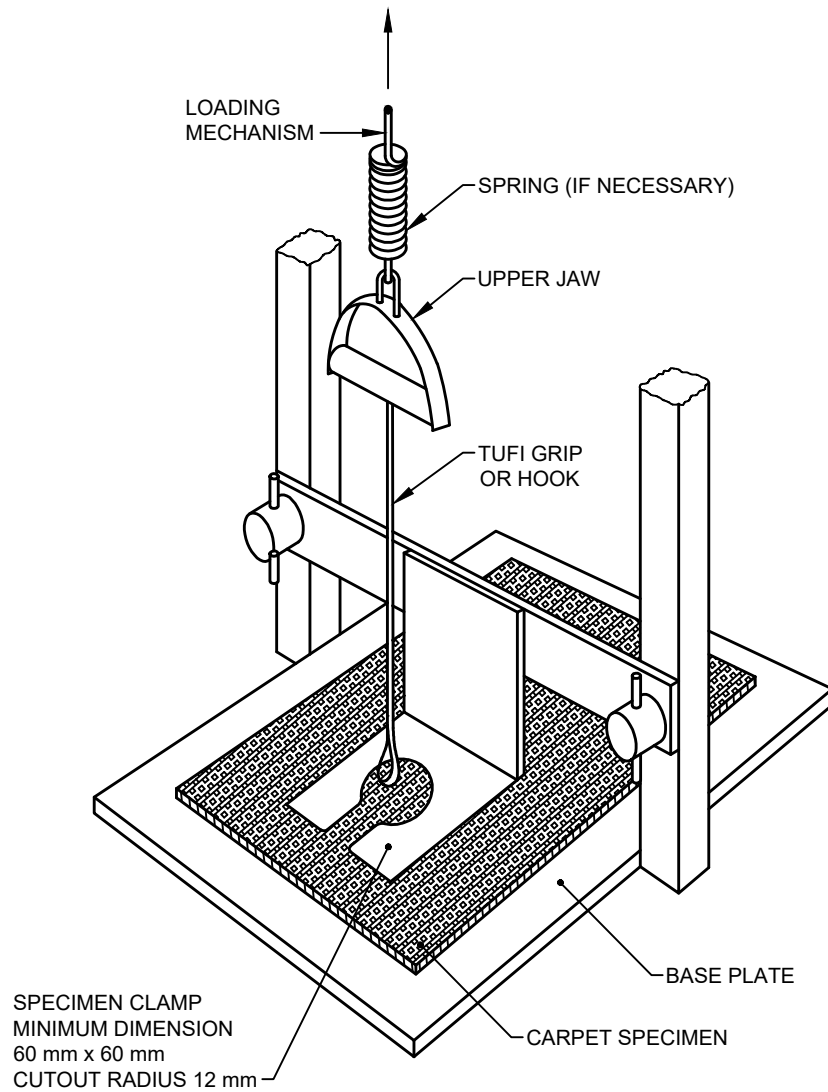


FIG. 6 TYPICAL FORM OF HORIZONTAL MOUNTING'

(TXD 04)

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